

Academic Integrity Intelligence Platform (AIIP)

Problem Statement

The rapid adoption of Artificial Intelligence in education has transformed how students prepare assignments. While AI tools improve productivity, they also introduce challenges such as AI-generated content, plagiarism, and improper citation practices. Traditional plagiarism detection systems primarily rely on exact text matching and often fail to identify paraphrased content, semantic similarities, or AI-assisted writing. Educational institutions need an intelligent, explainable, and scalable solution to assist faculty in evaluating assignments while promoting fairness and academic integrity.

Proposed Solution

The Academic Integrity Intelligence Platform (AIIP) is an AI-powered web application that leverages IBM Cloud and IBM watsonx.ai Foundation Models to analyze assignment submissions. Users upload assignment PDFs through a React-based interface. A FastAPI backend extracts text from the document and securely sends it to IBM watsonx.ai for analysis. The platform generates an Assignment Summary, Academic Integrity Score (0–100), Possible Plagiarism Observations, AI-Generated Content Analysis, Citation Suggestions, and Recommendations for Improvement.

Technologies Used

- IBM Cloud
- IBM watsonx.ai Foundation Models
- React.js, Vite, Tailwind CSS
- FastAPI (Python)
- PyPDF2
- IBM Identity and Access Management (IAM)

Expected Outcomes

- Reduce manual effort in assignment evaluation.
- Support educators with AI-assisted academic integrity analysis.
- Improve transparency through explainable reports.
- Encourage responsible use of AI in education.
- Provide a scalable foundation for institutional deployment.

Future Scope

- Integration with Learning Management Systems (Moodle, Canvas, Google Classroom).
- Advanced semantic plagiarism detection using vector databases.
- Multi-language assignment analysis.
- IBM watsonx Orchestrate integration for conversational academic assistance.
- Institutional analytics dashboards and scalable SaaS deployment on IBM Cloud.